# SNAP CIRCUITS

# SAFETY

RULE 1: No loose metal, don't let bare wires touch or it will short circuit.

RULE 2: battery in last, complete the circuit before turning it on.

RULE 3: If its hot - ST(STOP)P, If a component is warm or smells funny, disconnect the battery.

RULE 4: Snap, don't force, be gentle or it will break.

| | | |
|---|---|---|
| snap wire | # | connects to components and transports electricity |
| battery holder | B1 | batterys produce voltage which pushes electrons. |
| red LED | D1 | lights up |
| motor | M1 | converts electricity into motion. |
| 100Ω resistor | R1 | control how fast electricity flows |
| slide switch | S1 | connects (ON) or disconnect (OFF) the circuit. |
| base grid | | where you connect components and wires |
| 2.5v lamp | L1 | lights up |
| BUZZER | WC | vibrates to make noise |

Voltage gives the electrons the push it needs to flow. When electrons flow it is called current, similar to a water current. A circuit needs a complete loop to circulate the current. A switch can break the circuit, causing every thing to stop.

- I predict that adding a second LED will make both dimmer because the votage will be shared between both of them.

- photons are in inside of the LED causing it to light up.

# Troubleshooting Toolkit

- Check The Battery first
- Trace The Complete loop.
- Check polarity on LED's
- Verify all Snaps Are fully Seated.
- Simplify And test section.

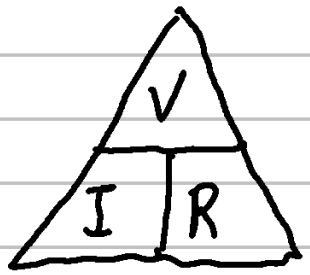
# Adding Motion

- In the motor circuit, chemical energy stored in the batteries is transformed into electrical energy.
- This electrical energy travels through the wires and reaches the motor, which converts it into mechanical energy.
- you can tell this is happening because the fan blade starts spinning as soon as you flip the switch.
- The circuit also releases a small amount of thermal energy, which you notice by carefully touching the motor after it runs for a while - it feels warm.
- When I added a resistor to the circuit, the motor slowed down because the resistor reduced the current, giving the motor less electrical energy to make the fan spin.

# Ohms Law $V = I \cdot R$

Voltage = Current · Resistance

Our battery is 3 volts, the LED is 0.02 amps (current)
Resistance?




ours: $R = \frac{V}{I}$ or $R = \frac{3}{0.02}$ $R = 150\Omega$

A $150\Omega$ resistor is perfect for protecting the led.

As I added more resistance it got dimmer untill it eventually went out

# The THRESHOLD Test

Which color will go out first, red yellow or green?

I perdict green will go out first because its already dim.

It went out: green, yellow, red.

The green light went out first because it has the highest voltage threshold, meaning its photons require more energy to be released. As the variable resistor increased resistance, it caused a significant energy drop in the circuit, quickly falling below the level needed to keep the green LED powered.

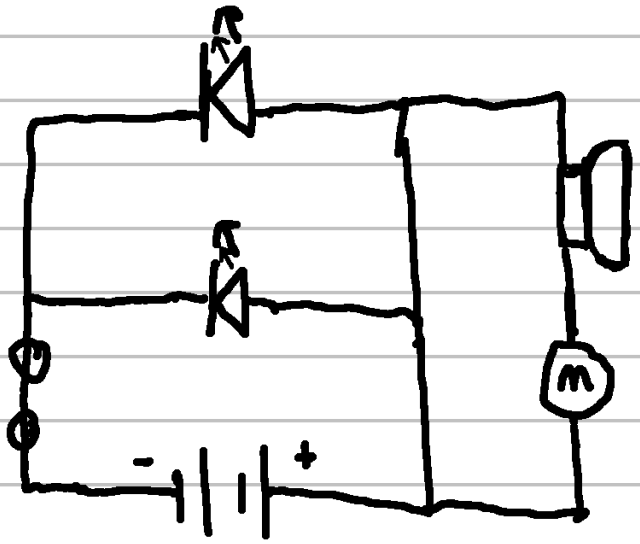
1. 5 different components

2. 2 led's

3. 1 parallel

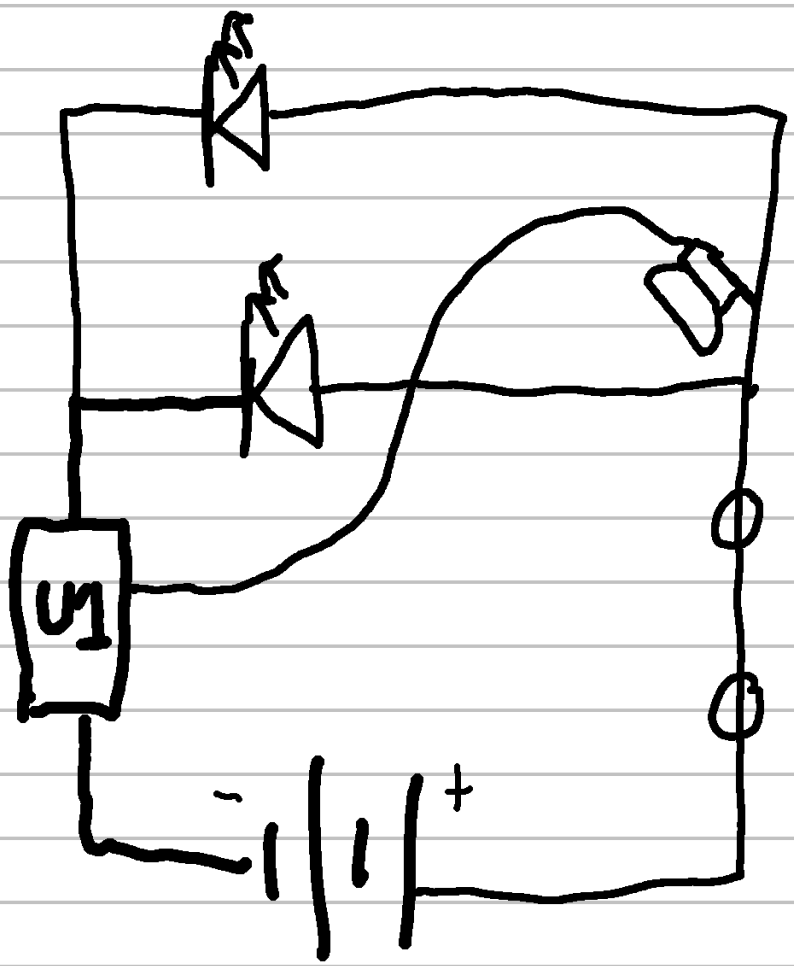
4. 2 transformation to energy from electrical

5. circuit diagram, list, why.




Didnt work

I think it didnt work because there were too many paths and the electrons didnt flow right

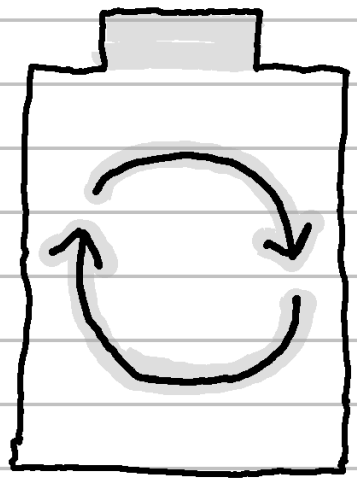



- U1: tells the buzzer what to play
- buzzer: makes noise
- switch; breaks and closes the circuit
- LED: lights up
- Battery: provides the push for electrons to start moving.

I made it a big parallel circuit so every thing could be on at once at its full potential. To fit all the components nicely I used 2 grids.

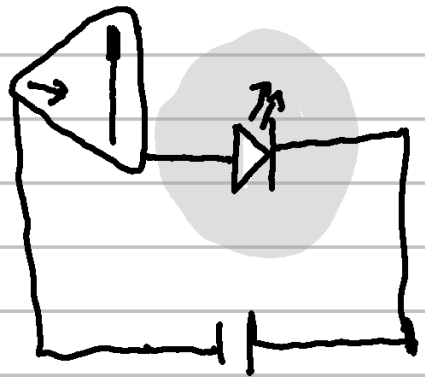
# The BIG 5

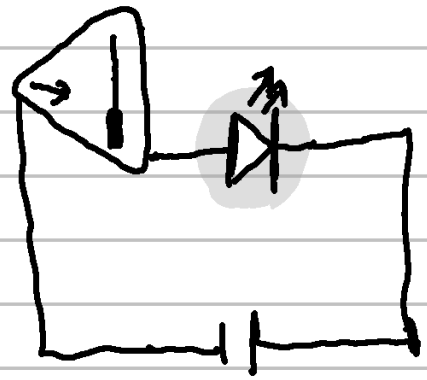
- For a circuit to flow it needs a power source to produce enough voltage to produce a current by pushing the electrons through the wires to go through the components and eventually reach the otherside of the power source in a complete loop.



- The difference between series and parallel circuits is that in a series circuit if one components is turned off all the rest go out because they are connected by one path. In a parallel circuit if one component is turned off the others stay on because there are multiple paths. In real life your house is a parallel circuit, if one light turns off your tv doesn't stop. A series circuit is like an old string of christmas lights, where if one burns out the rest turn off.

Ohm's law; $V = IR$, actually means voltage = current • resistance. When I used the variable resistor on Leds when I added more resistance the light got dimmer, it slowed the current to limit the voltage of the light.





Energy is tranformed in every component. In the battery its chemical energy, in the wires it turns into electrical energy, then it goes into the components.

- an LED turns into thermal energy (heat),
- a motor turns into mechanical energy (kinetic)
- a buzzer turns into sound waves (which comes from mechanical energy).

My top three most important steps in case my circuit doesn't work are:

1: Make sure the battery works and that they are fully secure.
2: Check if the loop is completly connected.
3: simplify the circuit down so its easier to find the error.